

When the total looks right but the stock still needs checking

A short inventory reporting case study on why aggregate accuracy does not always create operational confidence.

System OnHand

3.7K

System Available

2.1K

Operationally Available

2.1K

At first glance, the headline numbers look good:
The total stock that is available according to the system seems to be the same as the operationally available stock (stock on hand minus allocated in orders as well as quarantined stock).

When looking at the numbers individually, however, of the 24 products reviewed, 16 show a difference, with the variance range being substantial even if the net difference is only 9.

A negative difference (system availability > operational availability) can often be explained by quarantined stock. A positive difference, however, suggests there may be stock physically available that has not yet been reflected correctly in the system.

Largest Positive Difference

40

Largest Negative Difference

-30

SKUs Reviewed

24

SKUs with Difference

16

When adding the quantity in Quarantine, however, it becomes clear there is more to this difference.

Recommendation:

- Investigate whether all transactions have been entered into the system - missing deliveries or receipts affect the system totals
- Investigate time of physical count/report creation - a time difference could mean documents are no longer missing but not in one of the reports yet

SKU	System OnHand	System Available	Operationally Available	Difference	Quarantined Qty
B230	260	80	120	40	0
D420	160	105	135	30	0
E530	410	160	190	30	20
C310	180	120	145	25	0
A140	150	110	130	20	0
A110	85	65	80	15	0
D400	45	35	37	2	0
C330	130	90	85	-5	0
E510	140	70	65	-5	5
B210	110	95	89	-6	0
E540	125	85	78	-7	0
A130	64	52	44	-8	0
B240	72	52	44	-8	8
A120	200	125	115	-10	10
E500	75	55	45	-10	0
B220	95	95	83	-12	0
F600	60	45	33	-12	12
C320	90	65	50	-15	15
B200	300	180	155	-25	25
D410	220	120	90	-30	30
Total	2971	1804	1813	9	125

The total told me there was a difference. The next step is to find where that difference began.

Next week: tracing the largest variances back to stock movements and transaction timing.

The first step was to test the most likely explanations.

System availability is defined here as stock on hand less stock allocated to orders. Operational availability also removes stock held in quarantine.

A negative difference should therefore be explained by quarantined stock. Of the 13 negative differences identified, **7 reconcile exactly to the Quarantined Quantity**. The remaining **6 do not**, so they require further investigation. For positive differences, the first check was unposted receipts. **6 of the 7 differences can be explained**, including one SKU where an unposted receipt is partly offset by quarantined stock.

That leaves **7 SKUs in total that still don't reconcile from the information reviewed so far**.

SKU	Difference ▲	Quarantined Qty
D410	-30	30
B200	-25	25
C320	-15	15
B220	-12	0
F600	-12	12
A120	-10	10
E500	-10	0
A130	-8	0
B240	-8	8
E540	-7	0
B210	-6	0
C330	-5	0
E510	-5	5

SKU	Difference ▼	Quarantined Qty	Unposted Receipts
B230	40	0	40
D420	30	0	30
E530	30	20	50
C310	25	0	25
A140	20	0	20
A110	15	0	15
D400	2	0	0

The obvious explanations account for 9 of the 16 differences. The next step is to find out what is happening in the remaining 7.

Next week: tracing those exceptions back to physical counts, stock movements and transaction timing.

The next step was to look at the remaining exceptions

Having looked at quarantined quantities, the most likely reason for a difference between System Available Stock and Operational Available Stock, the difference for 7 items remained unexplained.

Investigating time of the stock count and time of issues and receipts by the warehouse then resolved four more differences. In each case, production stock has been issued operationally but the transaction had not yet been posted in the system. The next step was then to go back to the physical counts and compare quantities in the report with the counting sheets.

The counting sheets were then compared with the system counting records, including both quantities and Unit of Measure. This identified a different issue for B210: the system used 'each' as the standard UoM, while the counting sheet recorded 2 packs containing 3 items each.

A physical recount of E540 and D400 was performed and it was concluded errors were made in the original count.

SKU	Difference	Quarantined Qty	Unposted Issues	Unposted Receipts	ExceptionType
A130	-8	0	8	0	Production issue not posted
B220	-12	0	12	0	Production issue not posted
C330	-5	0	5	0	Production issue not posted
E500	-10	0	10	0	Production issue not posted
D400	2	0	0	0	Physical count surplus
B210	-6	0	0	0	Physical count shortage
E540	-7	0	0	0	Physical count shortage

SKU	Physical Count Difference	Exception Value	Stock Value	Impact (%) of Exception	ExceptionType
A130	-8	-195.20	1,561.60	12.50%	Production issue not posted
B220	-12	-154.80	1,225.50	12.63%	Production issue not posted
C330	-5	-184.50	4,797.00	3.85%	Production issue not posted
E500	-10	-96.00	720.00	13.33%	Production issue not posted
D400	2	0.84	18.90	4.44%	Physical count surplus
B210	-6	-42.60	781.00	5.45%	Physical count shortage
E540	-7	-55.30	987.50	5.60%	Physical count shortage

The exceptions were small in number, but their impact on the affected stock values ranged from 3.85% to 13.33%.

We now know what caused the remaining exceptions.

Next week: what this means for reporting confidence, process and where the controls should sit.

What the analysis tells us - and what should happen next

What we confirmed

The analysis started with a small net stock difference, but 16 of 24 reviewed SKUs did not reconcile individually. The investigation then worked through the differences in stages:

- expected differences were checked against quarantine and unposted receipts;
- remaining exceptions were traced through transaction timing and stock movements;
- unresolved items were checked against physical count records and Unit of Measure;
- the remaining discrepancies were confirmed as posting delays, counting issues or count-entry errors.

The main point is not simply that the differences were found. It is that they came from more than one part of the process.

What this means

The stock figures themselves were only part of the issue. The analysis suggests that reporting confidence also depends on:

- when warehouse transactions are posted;
- when physical stock counts take place;
- whether stock continues to move during a count;
- how quantities and Units of Measure are recorded;
- how quickly count results are entered and reconciled.

A technically correct report cannot remove uncertainty created earlier in the process.

Recommended next steps

Review the current stock-taking process from start to finish, with particular attention to timing and control. The areas I would examine next are:

- whether stock movements can be temporarily frozen while a count is taking place;
- whether all receipts, issues and transfers are posted before the count begins;
- whether count sheets use a standard format, including clear Unit of Measure;
- whether quantities are independently checked — for example, by using two people for selected or higher-value items;
- whether count results are entered into the system immediately after completion rather than being held offline;
- whether discrepancies are reviewed before normal stock movement resumes;
- whether higher-value or repeatedly problematic items need tighter controls or more frequent cycle counts.

Use independent verification where the value, risk or history of the item justifies it.

Use standardised count sheets with SKUs, location and Unit of Measure clearly shown, but consider hiding the expected quantity during the physical count.

SKU	Physical Count Difference	Unposted Issues	Exception Value	Stock Value	Impact (%) of Exception	ExceptionType
E500	-10	10	-96.00	720.00	13.33%	Production issue not posted
B220	-12	12	-154.80	1,225.50	12.63%	Production issue not posted
A130	-8	8	-195.20	1,561.60	12.50%	Production issue not posted
E540	-7	0	-55.30	987.50	5.60%	Physical count shortage
B210	-6	0	-42.60	781.00	5.45%	Physical count shortage
D400	2	0	0.84	18.90	4.44%	Physical count surplus
C330	-5	5	-184.50	4,797.00	3.85%	Production issue not posted

The analysis found the differences. The next piece of work is to reduce the number of opportunities for those differences to arise in the first place.

Good stock reporting does not begin with the dashboard. It begins with a process that gives the dashboard something reliable to report.